Table 1: Model variables and data sets. Every point is one Spectre transient of one comparison from reset (GPDK045, tt, LVT).

Variable / set	Range or size	Sampling / use
ΔV_{in} (input)	100 µV – 300 mV	log-uniform
VCM (input)	0.4 – 0.6 V	uniform
VDD (input)	0.9 – 1.1 V	uniform
T (input)	0 – 85 °C	uniform
t_d (output)	36 – 285 ps	clock 50 % to output VDD/2
E (output)	23 – 43 fJ	supply energy per cycle
train	2800	scrambled Sobol; training
val	600	uniform random; early stopping, sizing
test	600	uniform random; final accuracy (used once)
extrap	500	ΔV_{in} 1–100 µV, t_d 83–345 ps; used once
slices	306	9 conditions × 34 ΔV_{in} (1 µV – 300 mV); Figs 4, 8
reference curves	771	VCM = VDD/2, 9 conditions, 10 fV – VDD; Fig 9

Table 2: Structures of the frozen models (smallest network with training and validation RMS error $\leq 0.5\,\%$). Layers counted as in the course, input layer included.

	ReLU MLP	tanh MLP (control)	KBNN
Layers	3	3	4
Inputs	$\log_{10}\lvert\Delta V_{\text{in}}\rvert$, VCM, VDD, T	same	VCM, VDD, T; $\lvert\Delta V_{\text{in}}\rvert \to L, v$
Hidden layer	50 ReLU	12 tanh	2 blocks × 25 ReLU
Linear outputs	t_d, E	t_d, E	t_0, τ, E_0, s; g, q
Knowledge layer	–	–	$t_d = t_0 + \tau L + [g(v) - g(0)]$ $E = E_0 + sL + [q(v) - q(0)]$
Weights	352	86	381

All: relative-MSE loss over (t_d, E); full-batch Adam, 60 000 epochs, learning rate $3 \times 10^{-3} \to 3 \times 10^{-5}$ (cosine); best-validation weights; 5 seeds; RTX 5080, float32. KBNN: softplus on τ and s; $L = \ln(1\,\text{mV}/\lvert\Delta V_{\text{in}}\rvert)$, $v = \lvert\Delta V_{\text{in}}\rvert/0.3\,\text{V}$.

Table 3: Errors of the frozen models, |% error|, mean ± standard deviation over 5 seeds. The test and extrapolation sets were evaluated once, after the sizes were fixed.

Model	Set	t_d avg	t_d max	E avg	E max
ReLU MLP	training	0.40 ± 0.05	3.1 ± 0.8	0.13 ± 0.01	0.9 ± 0.1
	validation	0.43 ± 0.05	3.0 ± 0.7	0.13 ± 0.02	0.8 ± 0.1
	test	0.45 ± 0.06	6.0 ± 1.8	0.13 ± 0.01	0.9 ± 0.2
	extrap. (1–100 µV)	4.16 ± 0.92	16.4 ± 2.7	0.52 ± 0.23	2.4 ± 0.5
tanh MLP (control)	training	0.43 ± 0.02	4.0 ± 0.5	0.17 ± 0.04	1.4 ± 0.2
	validation	0.44 ± 0.02	3.9 ± 0.3	0.17 ± 0.03	1.1 ± 0.2
	test	0.44 ± 0.02	5.2 ± 0.3	0.17 ± 0.03	1.5 ± 0.3
	extrap. (1–100 µV)	3.64 ± 1.06	20.8 ± 4.5	2.73 ± 1.75	10.8 ± 5.1
KBNN	training	0.39 ± 0.04	2.6 ± 0.6	0.10 ± 0.01	1.3 ± 0.2
	validation	0.45 ± 0.04	2.8 ± 0.5	0.10 ± 0.01	0.6 ± 0.1
	test	0.45 ± 0.05	3.5 ± 0.8	0.10 ± 0.01	0.7 ± 0.2
	extrap. (1–100 µV)	0.61 ± 0.04	2.9 ± 0.2	0.18 ± 0.05	1.1 ± 0.2

Table 4: Training and evaluation cost (RTX 5080, float32). Epochs to a target: median over the 5 seeds trained together that reach it.

	ReLU MLP	tanh MLP	KBNN
Weights	352	86	381
Time per epoch (ms)	0.31	0.33	0.77
60k epochs, 1 / 5 networks (s)	18.7 / 18.7	19.2 / 19.8	44.8 / 46.4
Epochs to val. RMS 1 %	1 960	3 360	1 100
Epochs to val. RMS 0.5 %	11 000 (5/5)	17 500 (3/5)	7 700 (3/5)
Time to 0.5 % (s)	3.4	5.8	6.0
One prediction, CPU (µs)	9	9	45
Per prediction, batch of 10^5: CPU / GPU (ns)	66 / 1.3	7 / 0.5	51 / 2.3
Spectre, per comparison	$\approx$ 0.3 s		

Table 5: Cost of one metastability error rate at the nominal condition. A rate of 10^{-9} to ± 10 % needs about 10^{11} conversions by brute force.

Route	Cost
Brute-force Spectre (10^{12} comparisons)	$\approx 10^4$ years
GPU Monte Carlo with a model, 10^9 conversions	12 / 26 / 43 s (tanh / ReLU / KBNN)
Threshold sum with a model (1023 thresholds)	$<$ 1 ms
KBNN closed form	$\approx$ 5 µs
Threshold sum with Spectre delays (85 simulations)	$\approx$ 40 s of Spectre per condition

Table 6: Metastability of the 10-bit asynchronous SAR (sampling 25 % of the period, 200 ps overhead per comparison, VCM = VDD/2). Left: sampling rate for 10^{-9} errors per conversion, Spectre reference and model error in %. Right: each model's predicted rate divided by Spectre's, where Spectre gives 10^{-9} (0: the model predicts no errors). Median of 5 seeds.

VDD, T	$f_{\max}$ (MS/s)	KBNN %	ReLU %	tanh %	KBNN	ReLU	tanh
0.9 V, 0 °C	209.7	+0.5	−0.7	+5.3	0.38	3.6	0
0.9 V, 27 °C	209.1	+0.4	−1.1	+5.0	0.43	6.3	0
0.9 V, 85 °C	202.9	+0.3	−0.5	+5.3	0.57	2.5	0
1.0 V, 0 °C	255.3	−0.1	−5.3	−0.7	1.4	2.6×10^3	20
1.0 V, 27 °C	251.6	−0.2	−4.9	−0.1	1.8	1.7×10^3	2
1.0 V, 85 °C	242.2	−0.2	−4.2	+1.1	1.7	6.8×10^2	8.5×10^{-5}
1.1 V, 0 °C	284.5	+0.0	−5.7	−3.0	0.85	1.5×10^4	8.0×10^3
1.1 V, 27 °C	279.6	+0.1	−5.5	−2.5	0.71	9.9×10^3	5.1×10^3
1.1 V, 85 °C	269.1	−0.1	−5.0	−1.3	1.7	5.6×10^3	1.1×10^3